



भारतीय प्रौद्योगिकी संस्थान खड़गपुर
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Post Graduate Certificate in Data Analytics & Application of Generative AI

10 Months | Online Live Interactive Classes | Capstone Project



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#4 IN
INDIA

#215 IN THE
WORLD

Learn The Application of Generative AI in Data Analytics

Transformative Skills for The Next Leap in Your Career



Top Emerging Careers of The Data-Driven Era



Data Scientist :

Called the “S3Xiest Job of the 21st Century” by Harvard Business Review. Combines data analytics, machine learning & AI to extract insights and drive decisions.



Generative AI Specialist :

Focuses on building and applying AI systems that generate text, images, video, code & more. In-demand across industries like media, marketing, healthcare, finance and EdTech.



AI/ML Engineer :

Designs and deploys machine learning & deep learning models in real-world applications. Crucial for organizations moving toward automation and AI-driven solutions.



BI & Analytics Expert :

Translates raw data into dashboards and actionable strategies for business growth.



AI Product Manager :

Combines analytics with strategic vision to lead AI-driven product innovation.



Data / Analytics Engineer :

Builds and optimizes the pipelines that power advanced data analytics and AI applications.



11.5 Million Jobs by 2026 – The Big Data Boom :

According to the US Bureau of Labor Statistics, with 11.5 million jobs predicted by 2026 in big data, the demand for experienced professionals is already far outstripping the supply. People with analytics expertise are in high demand and market leaders such as Amazon, Google, Facebook (Meta) and Microsoft are offering huge salaries to talented professionals.

As a domain, big data analytics is moving fast. However, there are a core set of skills you need to kick-start a career in this field.

Source :- **nasscom**

Program Overview

The Post Graduate Certificate in Data Analytics and Generative AI is designed to equip learners with essential analytical, technical, and AI-driven creative skills required in today's data-centric world. The program combines core Data Analytics competencies with the rapidly emerging field of Generative AI, enabling learners to apply data-driven insights and AI-based innovation to solve complex business challenges.

Program Pedagogy



Live Faculty Sessions :

Engaging sessions led by industry experts



Applied Case Studies :

Business-focused, problem-solving examples.



Continuous Evaluation :

Ongoing assessment for steady progress.



Real-world Case Studies :

Learn from practical business scenarios.



Interactive Workshops :

Hands-on activities for deeper learning.



Capstone Project :

End-to-end project showcasing expertise.

Program Objectives

- Build strong foundations in data analysis, statistics and visualization.
- Provide hands-on exposure to Python, SQL and modern analytics tools.
- Develop expertise in machine learning and predictive analytics.
- Explore Generative AI concepts, tools and applications in real-world domains.
- Enable participants to complete capstone projects showcasing practical knowledge.

Why do an Online Program

An online program provides flexibility, global access & cost-effective learning while delivering industry-relevant skills through expert-led sessions and practical projects.

It empowers professionals to upskill, network & earn a valued certification without disrupting their careers.

- ✓ Flexibility
- ✓ Global Access
- ✓ Industry-Relevant Content
- ✓ Expert Faculty
- ✓ Cost-Effective
- ✓ Hands-on Learning
- ✓ Networking Opportunities
- ✓ Career Advancement
- ✓ Recorded Sessions
- ✓ Recognized Certification

About Program



120 + Hours

D2D Live Sessions



10 Months

Weekend Virtual Live Classes

Program Curriculum

Module 1 : (20 Hours) Foundations of Data Analytics

- Introduction to Data Analytics & Business Applications
- Basics of Data, Databases, and Data Processing
- Introduction to Python for Analytics
- SQL for Data Extraction and Management

Module 2 : (20 Hours) Statistics & Exploratory Data Analysis

- Descriptive & Inferential Statistics
- Probability & Distributions
- Hypothesis Testing
- Data Cleaning & Transformation
- Exploratory Data Analysis (EDA) with Python & Visualization

Module 3 : (20 Hours) Predictive Analytics & Machine Learning

- Analytics Maturity Framework
- Supervised Learning: Regression & Classification Techniques
- Unsupervised Learning: Clustering
- Model Evaluation & Validation – Confusion Matrix, MAPE
- Introduction to Deep Learning
- Real-world Case Studies – Identify Machine Learning Problems

Module 4 : (20 Hours) Generative AI Foundations

- Introduction to Artificial Intelligence & Generative AI
- Neural Networks, Transformers & Large Language Models (LLMs)
- Prompt Engineering for Text Generation
- Image, Audio & Video Generation Tools (DALL-E, Stable Diffusion, etc.)
- Ethics & Responsible AI Use

Module 5 : (20 Hours) Applied Generative AI

- Chatbots & Conversational AI
- AI-driven Content Creation (Text, Media, Code)
- Business: Marketing, Healthcare, Finance, Education
- Integrating Generative AI with Data Analytics Pipelines & Data Transformation
- Hands-on Use cases with OpenAI, Hugging Face, N8N & Emerging Platforms
- Fair practice for Applied Generative AI

Module 6 : (20 Hours) Capstone Project & Presentation

- End-to-End Data Analytics & Generative AI Project
- Problem Formulation & Data Collection
- Model selection, Building and Testing
- Model Insights & Visualization
- AI-Enhanced Solution Prototype
- Final Presentation & Evaluation

Note:

- Modules/ topics are indicative only, and the suggested time and sequence may be dropped/ modified/ adapted to fit the total programme hours. Case studies, real world examples and numerical illustrations are an integral part of multiple modules included in the course.
- The primary mode of learning for this programme is via live online sessions with faculty members. Post session video recordings will be made available until the programme duration.

Skills You will Gain

- ✓ Data Analytics Fundamentals
- ✓ Python Programming
- ✓ SQL for Data Management
- ✓ Data Statistics & Statistical Analysis
- ✓ Data Cleaning & Transformation
- ✓ Data Visualization & EDA
- ✓ Machine Learning Techniques
- ✓ Data Mining
- ✓ Data security
- ✓ Data On-Cloud
- ✓ Data Structure
- ✓ Data Presentation & Intergation
- ✓ Model Evaluation & Validation
- ✓ Deep Learning Foundations
- ✓ Generative AI Concepts
- ✓ AI-Driven Analytics
- ✓ Conversational AI Development
- ✓ AI Integration with Analytics
- ✓ Generative AI Integration
- ✓ Generative AI Tools & Design
- ✓ Agentic AI
- ✓ End-to-End Project Execution

Capstone Project

- ✓ Manufacturing
- ✓ IT
- ✓ Healthcare / Pharma
- ✓ Retail Industry

Program Certificate



Complete Career Process Support

Get end-to-end career guidance — from skill enhancement to job readiness. Our dedicated mentors, mock interviews and portfolio-building support ensure you're industry-ready.

✓ Resume Building Assistance:

Craft a professional, data-driven resume that highlights your analytics expertise and industry value.

✓ Guaranteed Interview Calls:

Get assured interview opportunities with top recruiters through our dedicated placement network.

✓ Diverse Job Roles:

Access multiple career tracks — from Data Analyst to Gen AI Strategist — across industries.

✓ Mentoring and Career Counselling:

Receive personalized guidance from industry experts to align your career path with market trends.

✓ Strong Job Market Demand:

Join one of the fastest-growing domains with soaring corporate demand for Data & AI talent.

✓ Career Advancement Opportunities:

Accelerate your professional growth with advanced analytics and Generative AI expertise.

Personal & Professional Development

✓ Higher Salaries:

Command premium compensation as organizations invest heavily in data-driven expertise.

✓ High-End Visualizations:

Transform complex data into impactful, decision-ready visuals using modern analytics tools.

✓ Industry Recognition:

Gain credibility and visibility through validated Data Analytics & Gen AI credentials.

✓ Integration:

Seamlessly connect AI tools with enterprise systems for smarter business outcomes.

✓ Built-in AI:

Leverage AI-powered analytics to automate, predict, and optimize with precision.

✓ Continuous Learning:

Stay ahead in the ever-evolving data and AI ecosystem with ongoing upskilling.

✓ Premium for Advanced Certifications:

Stand out in the job market with globally recognized, industry-valued certifications.

✓ Flexibility & Speed:

Deliver insights faster with adaptable, AI-powered analytics frameworks and automation.

✓ Expansive Community:

Join a thriving global network of analytics and AI professionals driving innovation.

✓ Quick Reporting:

Generate real-time insights and reports that accelerate strategic decision-making.

✓ Collaboration:

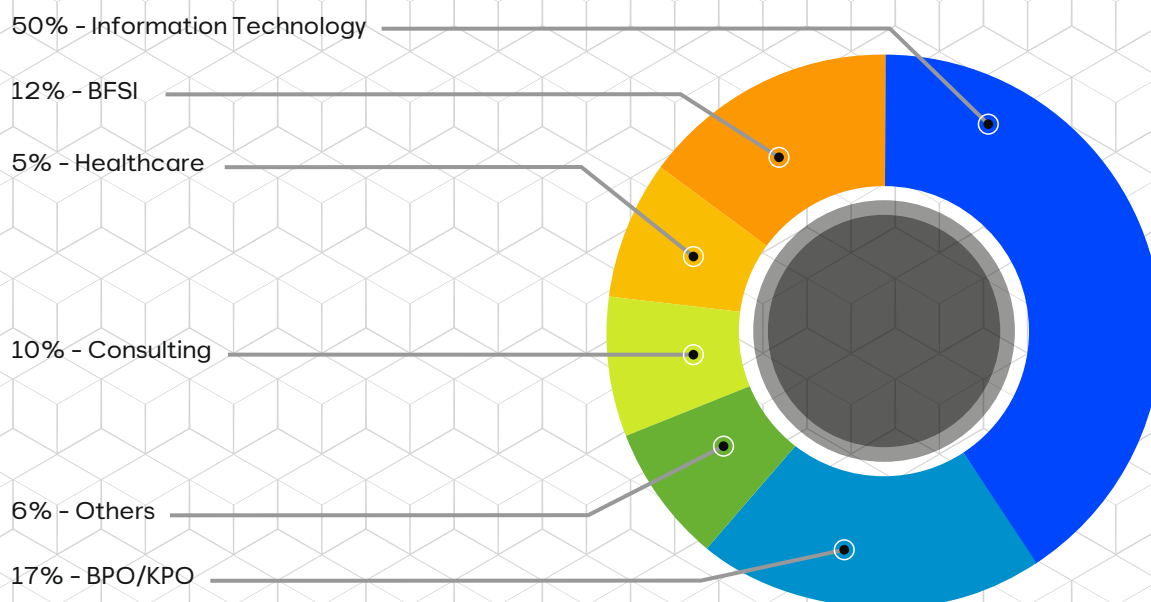
Empower cross-functional teams with shared, data-driven intelligence and unified dashboards.

✓ Global Reach:

Expand your professional footprint with skills in global-demand Data & Gen AI technologies.

Meet the Batch

Industries Our Learners Come From



Work Experience

Some of their current employers include



13%
12+ years

15%
9-12 years

25%
6-9 years

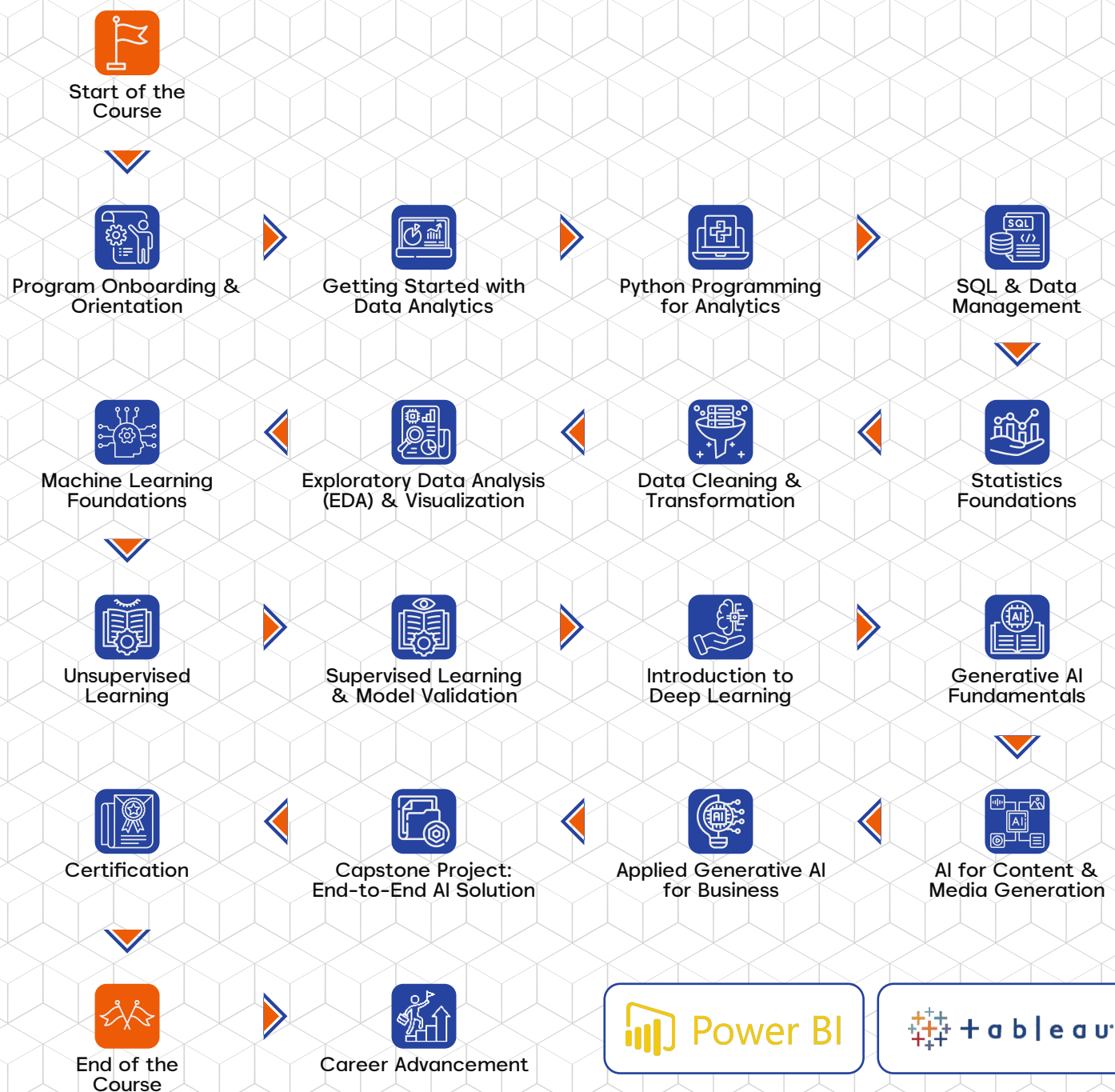
27%
3-6 years

20%
0-3 years



Learning Path

The learning path is structured to build strong foundations, advance through analytics and AI concepts and culminate in real-world applications. Each stage ensures progressive skill development with hands-on experience and guided expertise.



Note:

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Who can join the program?

- ✓ Graduate students aiming for AI/Analytics careers.
- ✓ Postgraduates enhancing data & AI skills.
- ✓ Working professionals seeking career growth.
- ✓ IT professionals moving into Data Science.
- ✓ Managers using data for decisions.
- ✓ Entrepreneurs exploring AI-driven innovation.
- ✓ Career switchers entering tech & AI roles.
- ✓ Researchers interested in Generative AI.

Registration Requirements

Graduation and Post-Graduation Education:

- Students may submit the marksheets up to the previous semester
- Working Professionals may submit their CV

ID Proof:

- Any Government-issued photo ID (PAN Card/ Driving License/ Passport, etc)

Registration Link & Process

Link : https://ktech.click/course_details/68b95b97fb9a0dab1b2bdfc5



Fill Online
Application



Profile
Shortlisting



Profile Acceptance
& Enrollment



You are now
a part of Program

Registration Fees

INR 2,000 + 18% GST

IIT Coordinator

Mrigank Sharad, Asst. Professor
mrigank@see.iitkgp.ac.in

Weekend Classes

Sunday: 10 AM to 1 PM

K-Tech Coordinator

Dr. Piya Sen : piya.sen@ktech.live
8584979746

Academic & Certificate Partner:



भारतीय प्रौद्योगिकी संस्थान खड़गपुर
INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

Outreach Partner:

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Refund & Cancellation Policy: ktech.click/refund_policies

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